Progress Report: Electromagnetic and Gravitational Wave Characterization of LISA binaries in the Extremely-Low Mass Survey

We have been modeling systems of double white dwarf (DWD) binaries in the Extremely-Low Mass (ELM) Survey with orbital periods $P < 1$ hr, since these will be resolvable, continuous gravitational-wave sources for the LISA mission at mHz frequencies. Below, we highlight three representative cases that span the range of light-curve types we encounter: a fully eclipsing system with the richest possible photometric information content, a system showing ellipsoidal variations as the dominant modulation, and a system with no clear photometric modulation at all, where the light curve alone cannot break the modeling degeneracies. All fits were performed with LCURVE, using an MCMC exploration of the posterior. The observed light curves, along with the best-fit model and best-fitted parameters, are shown in Section 1,2, and 3. We also show the characterization of these systems in the Gravitational Wave (GW) space by determining their GW amplitude and its uncertainties at the beginning of the mission and after 48 months of observations using the pipeline GLASS in Section 4.

1 System J0651+2844: the eclipsing benchmark

Of our sample, J0651+2844 has the shortest period, 12.75 min, making it the fastest system studied in this work. It is a fully eclipsing DWD, and its light curve simultaneously shows eclipses, ellipsoidal variations, and Doppler beaming, making it by far our best-case scenario.

Modeling this system with LCURVE, we obtain a very well-constrained solution using only three spectroscopic priors: the radial velocity semi-amplitude (K_2), effective temperature (T_2), and surface gravity ($\log g_2$) of the observed (ELM WD) component. Subscript 2 refers to the visible/observed WD, and subscript 1 to its unseen companion. The fit constrains the mass ratio $q = M_2/M_1$, the orbital inclination i, the scaled radii $r_{1,2} = R_{1,2}/a$, the velocity scale, and both effective temperatures. From those, we can derive physical parameters of the system, such as component masses $M_{1,2}$ and the orbital separation a. The results are shown in Figure 1 and in Table 1. The mass ratio, both masses, and the inclination are all recovered to within a few percent error. This is close to the best this kind of light-curve analysis can do.

To check whether an additional spectroscopic constraint could still improve this further, we reran the fit, adding a prior on the observed WD mass, M_2 (see Figure 2). We notice a small improvement of around 1.5-1.8 times smaller percentage errors for q, r_1, r_2, M_1, M_2, and a; while the inclination and temperatures remain almost unchanged. Here, both solutions already place every parameter at a percent-level precision because of the eclipse, which helps resolve the geometry of the system. Therefore, no real improvement is obtained from the extra prior.

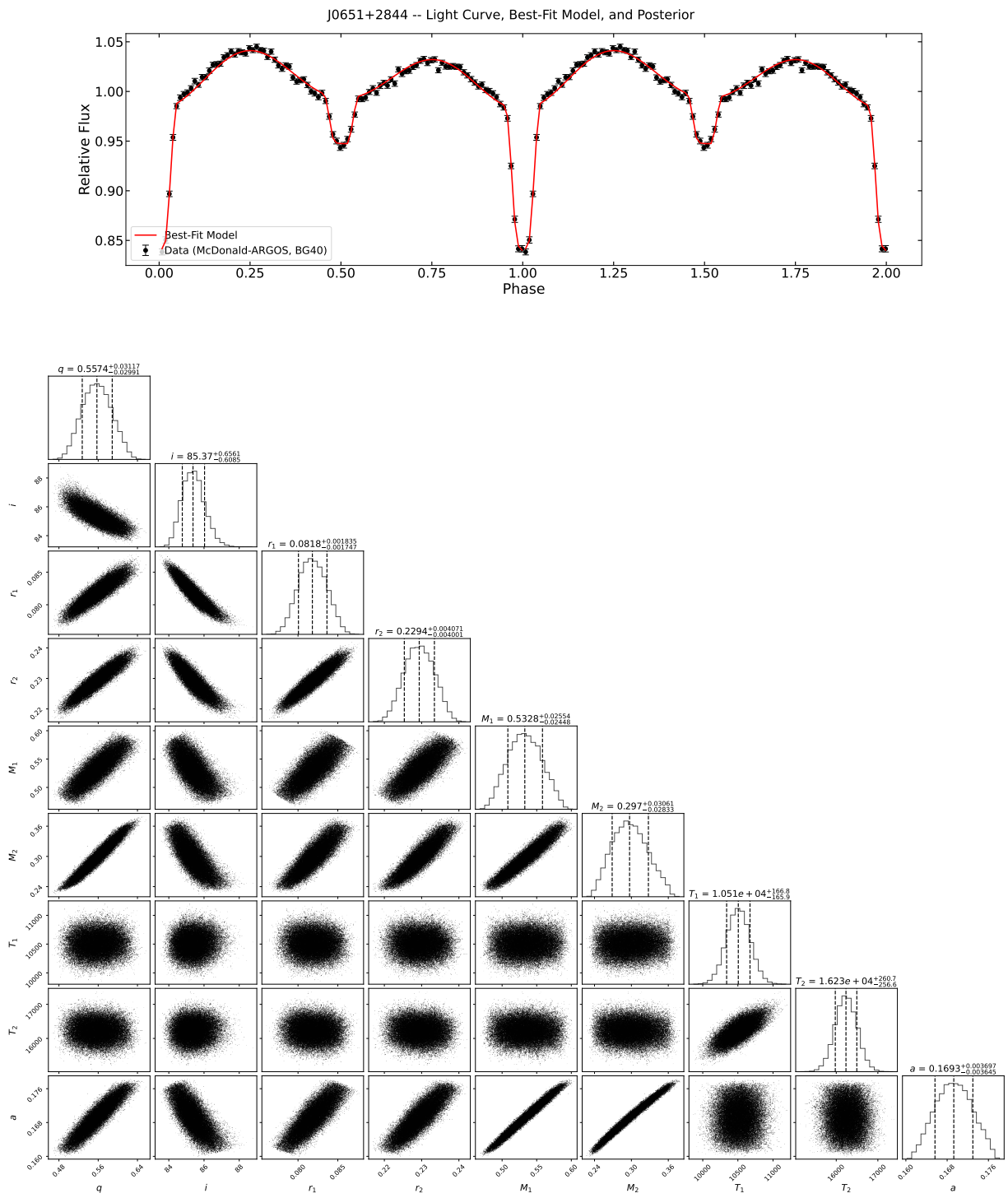


Figure 1: J0651+2844 ($P = 12.75$ min): phase-folded light curve with best-fit LCURVE model (top), and posterior corner plot for q, i, r_1, r_2, M_1, M_2, T_1, T_2, and a (bottom), using spectroscopic priors on T_2, K_2, and $\log g_2$.

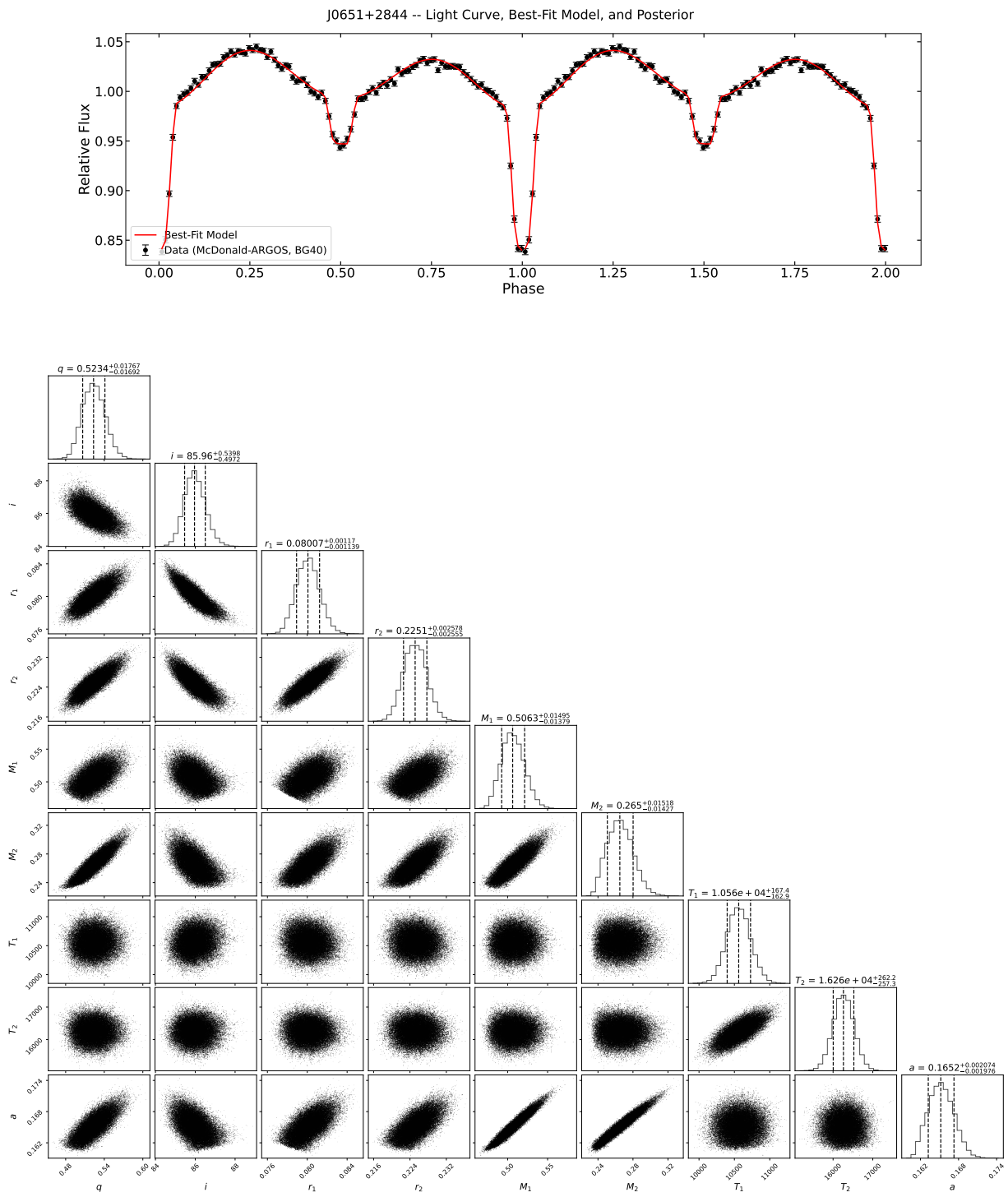


Figure 2: J0651+2844 ($P = 12.75$ min): phase-folded light curve with best-fit LCURVE model (top), and posterior corner plot for q, i, r_1, r_2, M_1, M_2, T_1, T_2, and a (bottom), using spectroscopic priors on T_2, K_2, $\log g_2$, and M_2.

2 Ellipsoidal variations systems: the case of system J0106–1000

A second, larger group of systems in our sample shows ellipsoidal variations as the dominant photometric modulation. These are intrinsically harder to constrain: without an eclipse to pin down the inclination and radii directly, the light curve alone carries less information, and the solution leans more heavily on the spectroscopic priors. Even so, we can still recover the inclination, mass ratio, and both masses at a useful level of precision, typically 10–25% for q and the two masses, and ∼10% for i, once a mass prior on the observed WD is included (see below).

We use J0106−1000 ($P = 39.1$ min) as the worked example. With the light curve plus the same three spectroscopic priors used before (T_2, K_2, $\log g_2$, see Figure 3), the mass ratio and masses come out substantially less precise than in the eclipsing case: q to $\sim 27\%$, M_1 to $\sim 23\%$, and M_2 to $\sim 48\%$ percentage errors. The inclination, by contrast, is already reasonably constrained by the ellipsoidal modulation alone with a percentage error of $\sim 12\%$.

Doing a similar test as before, we added a prior on the observed WD mass, M_2, to the LCURVE analysis (see Figure 4). This improves q to 12% (a ∼55% reduction in relative uncertainty) and M_2 to 11% (a ∼78% reduction, roughly 4–5× tighter), with M_1 improving similarly, to 12%. It does not, however, meaningfully improve the inclination, which is consistent with i being set by the amplitude and shape of the ellipsoidal modulation itself rather than by a mass prior. In both solutions, the temperature of the unseen component, T_1, remains essentially unconstrained by the light curve, at ∼72% relative uncertainty regardless of which prior set is used: with no eclipse, T_1 has no direct photometric handle, and is set almost entirely by whatever weak leverage the ellipsoidal light-curve shape provides through r_1.

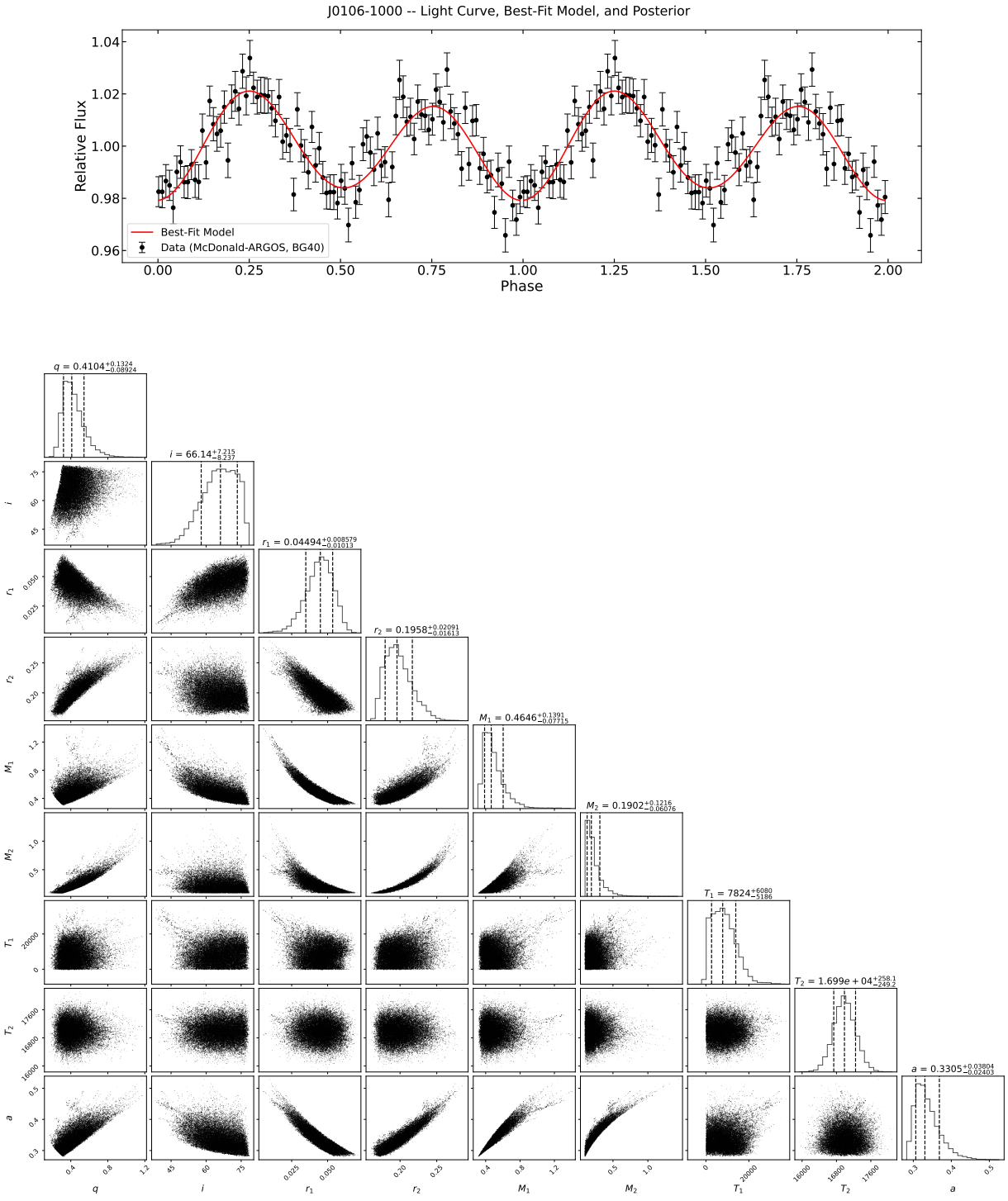


Figure 3: J0106−1000 ($P = 39.1$ min), baseline solution using priors on T_2, K_2, $\log g_2$ only.

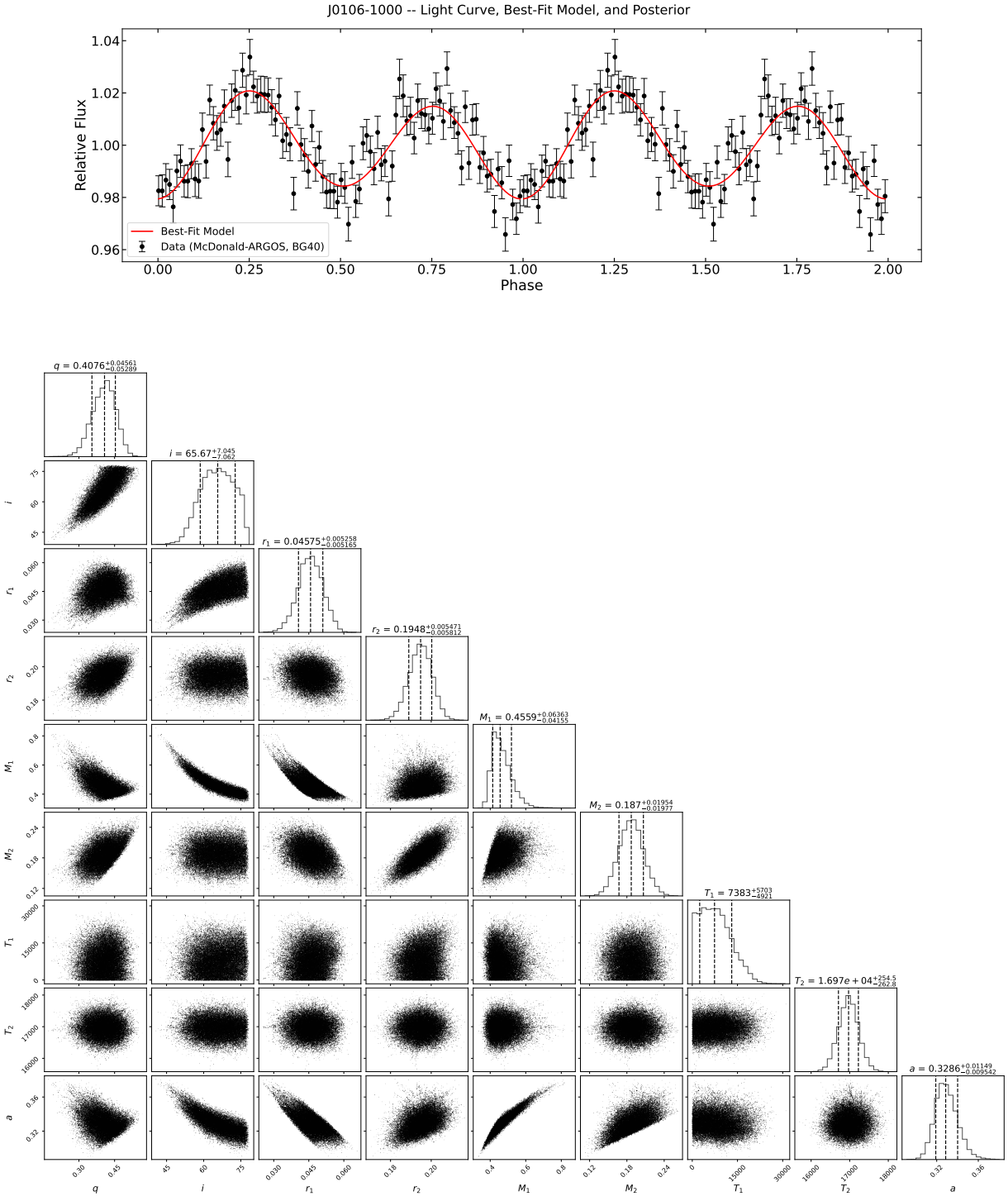


Figure 4: J0106−1000, with an additional spectroscopic prior on M_2. Compared to Fig. 3, q, M_1, and M_2 tighten considerably; i and T_1 do not.

3 Flat light curves: the case of system J2322+0509

Finally, a third group of systems shows no clear modulation at all in their light curves, which is itself diagnostic: both ellipsoidal and Doppler-beaming amplitudes scale strongly with $\sin i$ (and eclipses require i close to 90°), so a flat light curve points to a low orbital inclination. For these systems, a solution built on the light curve is unreliable. There is no real orbital-phase information in the data, and therefore, the posterior is dominated by whatever priors are used as input, not by the light curve itself.

We illustrate this with system J2322+0509 ($P = 20$ min), using the four spectroscopic priors setup (T_2, K_2, $\log g_2$, and M_2) as done for the ellipsoidal variation case. The inclination comes out unconstrained and strongly asymmetric, $i = 24.9^{+20.8}_{-6.0}$ deg (upper percentage uncertainty $\sim 84\%$), and the corner plot shows a clear degeneracy (see Figure 5). The same lack of constraint carries over to q (67% error), r_1 (33% error), r_2 (14% error), T_1 (71% error), and M_1 (52% error). By contrast, M_2 (7% error) and T_2 (2% error) come out tightly constrained, but only because they are driven by the input priors themselves, not because the light curve constrains them. Breaking degeneracies for this class of systems will require an independent constraint beyond what a light-curve solution can supply on its own. Barrientos et al. (2025) is a good example of how to attack this type of system.

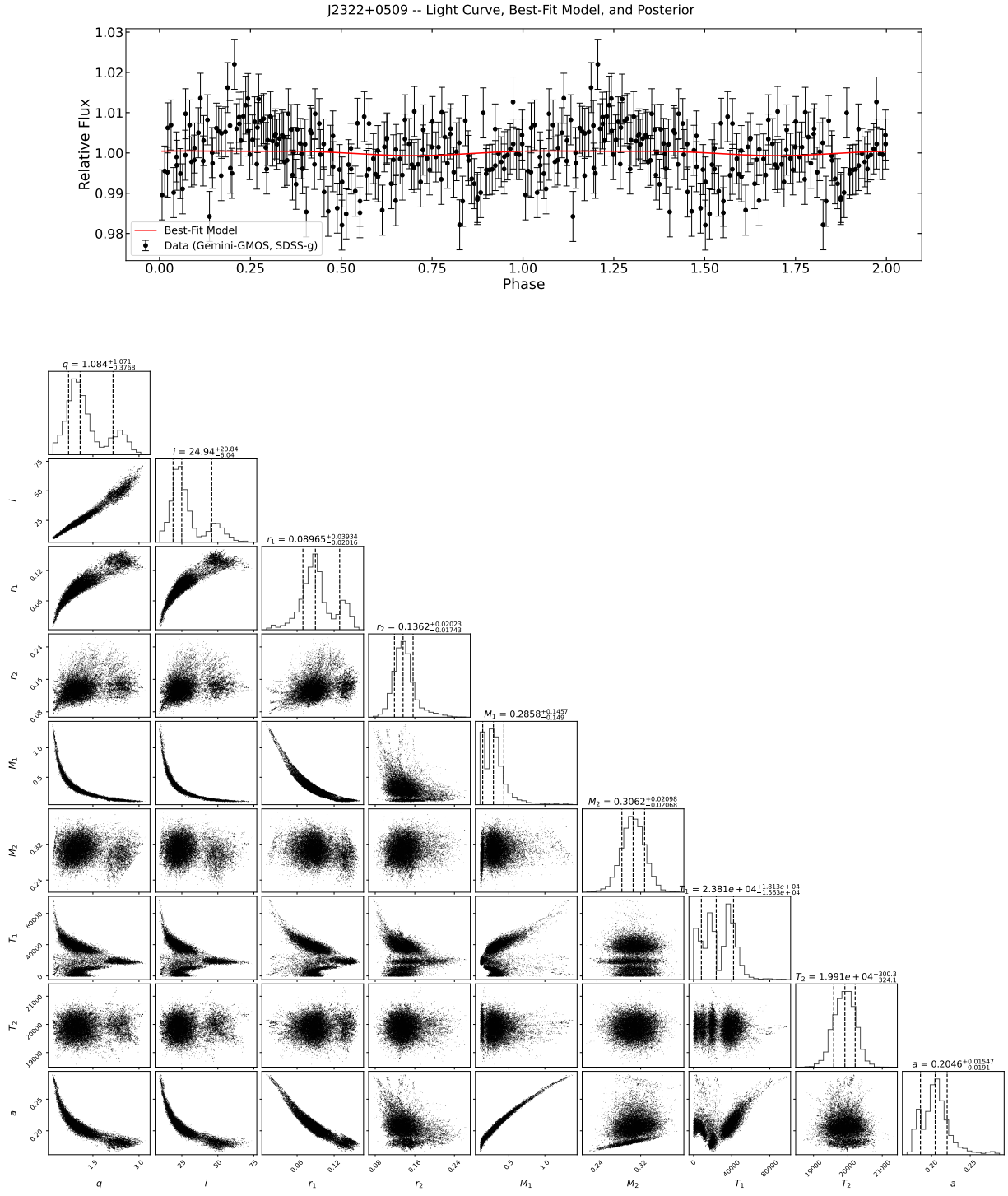


Figure 5: J2322+0509 ($P = 20$ min): flat light curve and degenerate posterior, using priors on T_2, K_2, $\log g_2$, and M_2.

Table 1: Posterior median parameters (16th/84th percentile uncertainties) for the three highlighted systems. Subscript 2 denotes the observed (ELM) component, subscript 1 its unseen companion; r_1, r_2 are radii scaled by the binary separation a (in $R_\odot$); M_1, M_2 in $M_\odot$; T_1, T_2 in K; i in degrees. "Priors" lists which spectroscopic constraints were used in each fit.

System	Priors	q	i (deg)	r_1	r_2	M_1 ($M_\odot$)	M_2 ($M_\odot$)	T_1 (K)	T_2 (K)	a ($R_\odot$)
J0651+2844	$T_2, K_2, \log g_2$	$0.557^{+0.031}_{-0.030}$	$85.4^{+0.7}_{-0.6}$	$0.0818^{+0.0018}_{-0.0017}$	$0.2294^{+0.0041}_{-0.0040}$	$0.533^{+0.026}_{-0.024}$	$0.297^{+0.031}_{-0.028}$	10510^{+170}_{-170}	16230^{+260}_{-260}	$0.1693^{+0.0037}_{-0.0036}$
J0651+2844	$T_2, K_2, \log g_2, M_2$	$0.523^{+0.018}_{-0.017}$	$86.0^{+0.5}_{-0.5}$	$0.0801^{+0.0012}_{-0.0011}$	$0.2251^{+0.0026}_{-0.0026}$	$0.506^{+0.015}_{-0.014}$	$0.265^{+0.015}_{-0.014}$	10560^{+170}_{-160}	16260^{+260}_{-260}	$0.1652^{+0.0021}_{-0.0020}$
J0106−1000	$T_2, K_2, \log g_2$	$0.410^{+0.132}_{-0.089}$	$66.1^{+7.2}_{-8.2}$	$0.0449^{+0.0086}_{-0.0101}$	$0.196^{+0.021}_{-0.016}$	$0.465^{+0.139}_{-0.077}$	$0.190^{+0.122}_{-0.061}$	7820^{+6080}_{-5190}	16990^{+260}_{-250}	$0.331^{+0.038}_{-0.024}$
J0106−1000	$T_2, K_2, \log g_2, M_2$	$0.408^{+0.046}_{-0.053}$	$65.7^{+7.0}_{-7.1}$	$0.0458^{+0.0053}_{-0.0052}$	$0.195^{+0.005}_{-0.006}$	$0.456^{+0.064}_{-0.042}$	$0.187^{+0.020}_{-0.020}$	7380^{+5700}_{-4920}	16970^{+250}_{-260}	$0.329^{+0.011}_{-0.010}$
J2322+0509	$T_2, K_2, \log g_2, M_2$	$1.08^{+1.07}_{-0.38}$	$24.9^{+20.8}_{-6.0}$	$0.090^{+0.039}_{-0.020}$	$0.136^{+0.020}_{-0.017}$	$0.29^{+0.15}_{-0.15}$	$0.306^{+0.021}_{-0.021}$	23810^{+18130}_{-15630}	19910^{+300}_{-320}	$0.205^{+0.015}_{-0.019}$

4 Forecasting LISA detectability: gravitational wave signal-to-Noise ratio

Beyond the EM (light-curve) modeling above, we also want to know whether, and on what timescale, LISA would actually detect each of these systems as a gravitational-wave source. We forecast this with the `GLASS` package and its `vgb_mcmc` sampler, which models the expected GW signal from a known galactic binary given its orbital and physical parameters. For each source, we build an injection file with fixed GW frequency, frequency derivative, sky position, and component masses (all set to their best-fit/catalog values), while the inclination and distances are instead drawn from a Gaussian prior set coming from the EM constraints from LCURVE and Gaia distances, respectively. These are propagated directly into the GW forecast, rather than fixing them at a single point estimate. The simulated data include stationary Gaussian instrument noise following the LISA Data Challenge 2a (LDC2a) noise model, together with the astrophysical foreground produced by the unresolved Galactic DWD population, which is the dominant confusion-noise term in this frequency range.

We simulated two LISA observation baselines, 3 months (an early-mission checkpoint) and 48 months (approximately the nominal mission duration), and computed the recovered SNR posterior (median and 16th–84th percentile range) for each system (Fig. 6). Adopting a detection threshold of SNR $= 6$, J0651+2844 is already detectable after just 3 months (SNR ≈ 8), and its SNR grows dramatically over the mission, reaching a median of ≈ 105 by 48 months, making it one of the most promising LISA verification sources in this sample. J0106−1000 and J2322+0509 are far worse at 3 months, both falling well below the SNR $= 6$ threshold. By 48 months, J2322+0509 crosses into detectable territory (SNR ≈ 8), while J0106−1000 remains below the detection limit for the full 48-month baseline (SNR ≈ 3–4), consistent with it being both the intrinsically fainter GW source of the two.

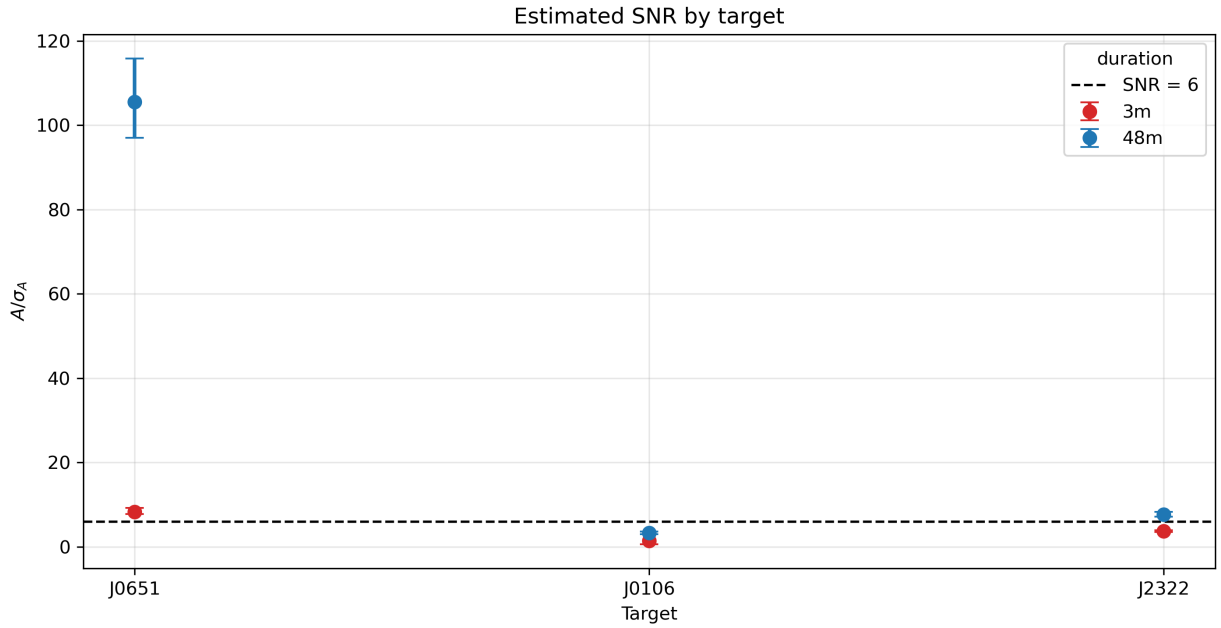


Figure 6: Forecast LISA signal-to-noise ratio for each system after 3 months (red) and 48 months (blue) of simulated observation, from `GLASS/vgb_mcmc` injections using inclination and distance priors drawn from the LCURVE (EM) posteriors and Gaia distances, respectively. Points show the median recovered SNR; error bars span the 16th -84th percentile range. The dashed line marks the adopted detection threshold, SNR = 6.